November 5, 2025 6:00 PM

Practice

1. A grey squirrel drops a 0.02 kg walnut from a branch that is 8.0 m high. Determine the change in potential energy of the walnut between the branch and the ground. [ans: 1.6 J]
2. The weightlifter in Sample Problem 2 increases the mass of the weights on the barbell and lifts it one more time. The new lift increases the potential energy of the bar by 660 J. Calculate the new mass of the barbell. [ans: 31 kg]
3. Calculate the work required by the student in Sample Problem 3 to stack 2 more physics books on the original stack of 10. [ans: 12 J]

1) $E_g = mg\Delta y$

$\Delta E_g = 0.02(9.8)(8) = 1.568\ J$

2) $E_g = 660\ J$

$\Delta y = 2.2\ m$

$mg = \frac{660}{2.2}$

$m = \frac{300}{9.8}$

$m = 30.6\ kg$

$= 31\ kg$

3) Δy of 1 book $= 3.6\ cm = 0.036\ m$

m of 1 book $= 1.6\ kg$

0+1+2+3+4+5+6+7+8+9

~~10~~ 5 m

10 book equation $= (1.6)(9.8)[(0.036)(45)]$

$= 25\ J$

12 book $= (1.6)(9.8)[(0.036)(0+1+2+3+4+5+6+7+8+9+10+11)]$

$= (1.6)(9.8)(0.036)(66)$

$= 37.255$

$W = E_{g_2} - E_{g_1}$

$= 37.255 - 25$

$= 12.25\ J$

1. A 2.5 kg piece of wood falls onto a carpenter's table from a height of 2.0 m above the table.
   (a) Calculate the kinetic energy of the wood as it hits the table.
   (b) Calculate the speed of the wood as it hits the table.
2. Calculate the gravitational potential energy relative to the ground of a 5.0 kg Canada goose flying at a height of 553 m above the ground.
3. A hockey referee drops a 175 g hockey puck from rest vertically downward from a height of 1.05 m above the ice surface.
   (a) Determine the gravitational potential energy of the puck relative to the ice before the referee drops it.
   (b) Calculate the change in gravitational potential energy of the puck as it drops from the referee's hand to the ice surface.
   (c) Calculate the work done on the puck by gravity as the puck travels from the referee's hand to the ice surface.
4. You lift your pet cat vertically by 2.0 m, and then you lower it vertically by 2.0 m. During this exercise, is the total work done by gravity positive, negative, or zero? Explain your answer.
5. A pole vaulter clears the bar at a height of 5.4 m, and then falls to the safety mat. The change in the pole vaulter's gravitational potential energy from the bar to the mat is -3.1×10^3 J. Calculate the pole vaulter's mass.
6. A 0.46 kg golf ball on a tee is struck by a golf club. The golf ball reaches a maximum height where its gravitational potential energy has increased by 155 J from the tee. Determine the ball's maximum height above the tee.
7. A 59 kg snowboarder descends a 1.3 km ski hill from the top of a mountain to the base (**Figure 4**). The slope is at an angle of 14° to the horizontal. Determine the snowboarder's gravitational potential energy relative to the mountain base when she is at the top.

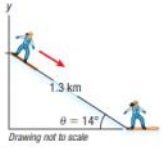


Figure 4

8. Suppose that you have N identical boxes in your room, each with mass m and height Δy. You stack the boxes in a vertical pile.
   (a) Determine the work done to raise the last box to the top of the pile. Express your answer in terms of the variables m, g, N, and Δy.
   (b) Determine the gravitational potential energy, in terms of m, g, N, and Δy, that is stored in the entire pile.
9. A gallon of gas contains about 1.3×10^8 J of chemical potential energy. Determine how many joules of chemical potential energy are stored in each litre of gas (3.79 L = 1 gallon). Calculate the height that the amount of chemical potential energy in 1 L of gas could raise all the students in your class if it was all converted to gravitational potential energy. You will have to make assumptions about the mass of the students. State your assumptions and show your calculations.

1) $E_K = \frac{mv^2}{2}$

$E_g = mg\Delta y$

$E_K = E_g$

a) $E_K = 2.5(9.8)2$

$= 49\ J$

$49 = \frac{mv^2}{2}$

$\sqrt{\frac{49(2)}{2.5}} = v$

b) $v = 6.26\ m/s$

2) $E_g = (5)(553)(9.8)$

$= 27\ 097\ J$

3) $m = 0.175\ kg$

a) $E_{g_1} = 0.175(-9.8)(-1.05)$

$= 1.8\ J$

b) $\Delta E_g = E_{g_2} - E_{g_1}$

$= 0 - 1.8$

$\Delta E_g = -1.8$

c) $W = -1.8$

$W = \Delta E_g$

4) $\Delta d = 0$

so work done is zero

5) $E_g = 3.1 \times 10^3$

$3.1 \times 10^3 = (-5.4)(-9.8)(m)$

$m = 58.6\ kg$

6) $m = 0.46$

$E_g = 155\ J$

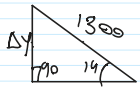
$\frac{155}{(-9.8)} = -15.816$

$\frac{-15.816}{0.46} = -34.383$

Max height $= 34.383\ m$

7) ☆

$mg\Delta y$



$\sin 14 = \frac{\Delta y}{1300}$

$1300 \sin 14 = \Delta y$

$\Delta y = 314.498\ m$

$E_g = (-314.5)(-9.8)(59)$

$= 181\ 843\ J$

$E_k = E_g$

$\frac{mv^2}{2} = 181843$

$v = \sqrt{\frac{181843 \times 2}{59}}$

- first box doesn't move so no work
- second box is lifted Δy. third is lifted $2\Delta y$
- Last box $= mg(N-1)\Delta y$

b) BS answer

check Mr Panchbhaya's website,

terms of the variables m, g, N, and Δy.

(b) Determine the gravitational potential energy, in terms of m, g, N, and Δy, that is stored in the entire pile.

9. A gallon of gas contains about 1.3×10^8 J of chemical potential energy. Determine how many joules of chemical potential energy are stored in each litre of gas (3.79 L = 1 gallon). Calculate the height that the amount of chemical potential energy in 1 L of gas could raise all the students in your class if it was all converted to gravitational potential energy. You will have to make assumptions about the mass of the students. State your assumptions and show your calculations.

- Last box = $mg(N-1)\Delta y$

b) BS answer
check Mr Panchbhaya's website,

a) 1 gallon = 1.3×10^8 J

3.79 L = 1.3×10^8 J

1 L = 3.43×10^7 J

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